

## 5<sup>th</sup> Set of Problems

1. Consider the robotic manipulator illustrated in Figure 1.
  - (a) According to the Denavit-Hartenberg methodology, illustrate the reference frames and build the corresponding table of parameters.
  - (b) Solve the manipulator's direct kinematics.
  - (c) Express the manipulator's direct kinematics with respect to the illustrated base frame, which is arbitrarily displaced from the origin of frame zero.
  - (d) Determine the end-effector's geometric Jacobian, to express its linear and angular velocities, with respect to frame zero.
  - (e) Express the obtained velocities with respect to the base frame.

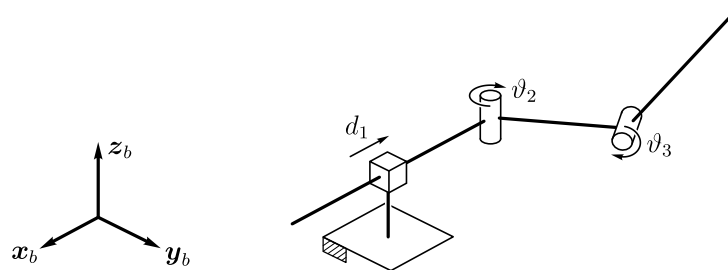


Figure 1: Stadium-toroidal manipulator.

2. Consider the aerial platform illustrated in Figure 2. Neglect the basket's rotation about the telescopic arm, so that the end-effector  $e$  corresponds to the telescopic arm's extremity. Assume also that the truck is unmovable.
  - (a) According to the Denavit-Hartenberg methodology, illustrate the reference frames over an equivalent joint diagram, and build the table of parameters.
  - (b) Determine the end-effector's geometric Jacobian.

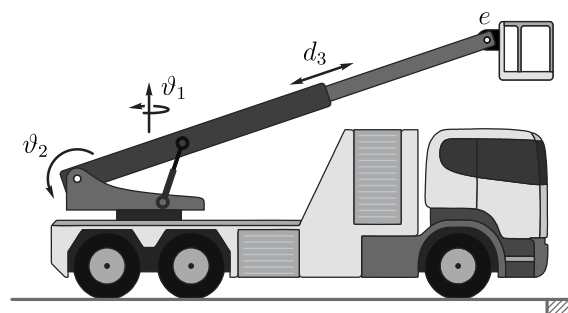


Figure 2: Fire engine with aerial platform.

3. Consider the robotic manipulator illustrated in Figure 3.
- Determine the manipulator's geometric Jacobian, at the end-effector.
  - Determine the end-effector's linear velocity and angular velocity.
  - Determine the end-effector's linear acceleration and angular acceleration.
  - Express the linear acceleration in the end-effector's frame of reference.
  - Given a desired velocity for the end-effector  $(\dot{\mathbf{r}}_e)_d = (\dot{e}_x \ \dot{e}_y \ 0)^T$ , determine the joints' rate of change,  $\dot{\mathbf{q}}$ , which achieve such task.

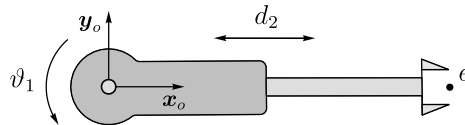


Figure 3: Polar manipulator.

4. [Computer Aided Problem] Consider the manipulator illustrated in Figure 4.
- Determine the end-effector's geometric Jacobian and the manipulator's internal singularities.
  - Given the exertion of an arbitrary generalized force at the end-effector, written as  $\gamma_e^o = (f_x \ f_y \ f_z \ \mu_x \ \mu_y \ \mu_z)^T$ , determine the corresponding joint torques,  $\tau$ , across the manipulator.

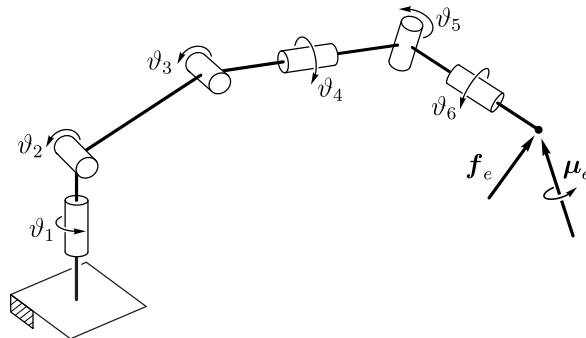


Figure 4: Anthropomorphic manipulator with spherical wrist.